



School of Engineering
Jawaharlal Nehru University

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SOUVENIR



PREFACE

We are pleased to present the E-Souvenir of ISoE 2025, International Symposium on Engineering, featuring the abstracts of all participating students and researchers. This compilation reflects the diversity of ideas, innovation, and academic enthusiasm within the School of Engineering, Jawaharlal Nehru University. The symposium aims to offer a platform for young engineers to share their work, exchange perspectives, and explore emerging trends across disciplines. We hope this souvenir serves as both a record of the event and an inspiration for future research and collaboration.

We extend our sincere appreciation to the participants, reviewers, faculty coordinators, and organizing team for their contributions to the ISoE 2025.

About SoE

The School of Engineering, Jawaharlal Nehru University (JNU), New Delhi, established in 2018, is a rapidly growing center of excellence dedicated to cutting-edge research and innovation in engineering and technology. With academic programs spanning Computer Science and Engineering, Electronics and Communication Engineering, and Mechanical Engineering, the School offers B.Tech, M.Tech, and Ph.D. programs that blend strong theoretical foundations with practical applications.

Note

I am delighted to share the E-Souvenir of ISoE 2025, which showcases the creativity and scientific commitment of our engineering community. The abstracts included here represent the curiosity, dedication, and problem-solving spirit that we strive to cultivate at the School of Engineering.

I congratulate all contributors and commend the organizing team for their efforts. May this souvenir encourage continued innovation, collaboration, and excellence.

Dean

School of Engineering,

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A Study on the influence of the Integration of Indian Knowledge System (IKS) into the School System in Mathematics

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The objective of this study is to integrate the Indian Knowledge System (IKS) into the mathematics curriculum of schools, aiming to enrich it by acknowledging India's intellectual legacy, cultural, and spiritual traditions. This integration is not intended to replace existing methods; rather, by utilizing effective Vedic computational techniques, educators can make mathematics more engaging, relevant, and powerful for future generations of learners. The study covers essential concepts such as algebra, geometry, numeracy, the decimal numeral system, the concept of zero, and various systematic arithmetic principles. Notable classical scholars like Aryabhata, Brahmagupta, and Bhaskara II contributed significantly to advancements in trigonometry, quadratic equations, and indeterminate solutions. IKS is recognized as one of the oldest and most impactful traditions globally, having developed over more than three thousand years. Recent research and the National Education Policy (NEP) 2020 highlight the contemporary relevance and integration of Indigenous Knowledge Systems (IKS) in mathematics education, focusing on the efficiency and speed of Vedic sutras (aphorisms) in mental arithmetic and problem-solving, as well as their interdisciplinary links to Astronomy and Linguistics, which promote computational logic and algorithmic reasoning. Currently, IKS is the subject of active research and advocacy as a comprehensive, culturally rooted, and rigorous source of mathematical knowledge, aimed at enhancing current teaching practices and computational problem-solving strategies.

Keywords:

Vedic Mathematics, Baudhāyana Theorem, Sulba Sutras, Mathematical History, Vedic Rituals, Dharma Sūtras, The Indian Knowledge System.

Transfer Learning-Based Approach for Hindi Text Summarization Automation

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Text summarization helps in reducing large volumes of text into concise and meaningful summaries, which is especially important in multilingual settings. Although a lot of progress has been made for English and other high resource languages, the Hindi language still faces many challenges such as limited high-quality datasets, fewer evaluation resources, and the absence of effective methods designed specifically for Hindi. Existing multilingual models like mT5 and IndicBART provide basic summarization support, but they often fail to handle the complex structure and free word order of Hindi sentences, leading to loss of important information and weak clarity in summaries. The method also considers multi-document summarization, where information from multiple related Hindi documents must be combined into a well-structured summary, which is challenging due to dialectal variations and code-mixed content. In addition, most current evaluation methods depend on automatic metrics like ROUGE and BLEU, which mainly check word overlap and cannot fully judge the meaning or factual correctness of summaries. To handle this limitation, this study will combine automatic metrics with human evaluation to ensure a more complete and accurate assessment of summary quality. Additionally, there is limited exploration of cross lingual transfer strategies where high-resource language models are effectively adapted to Hindi through controlled fine-tuning or knowledge refining. To address these challenges, the present research aims to develop an automated Hindi text summarization method using transfer learning techniques. The method will focus on improving the contextual understanding and factual correctness of generated summaries while dealing with data limitations through effective model adaptation. The work will also emphasize dataset preparation, fine-tuning of models, and reliable evaluation strategies to build a practical and linguistically sound summarization method suitable for real world applications.

Keywords: Hindi Text Summarization, Indic Languages, Transfer Learning, Low-Resource Languages, Abstractive Summarization

Automated Plant Angle Extraction from Multi-View Mustard Dataset

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Plant phenotyping represents a critical bottleneck in precision agriculture and crop improvement programs, where manual measurement of structural traits is labor-intensive, time-consuming, and subjective. With thousands of plant genotypes requiring phenotypic evaluation across multiple growth stages, automated computational methods have become essential for high-throughput breeding programs. Traditional manual measurement approaches struggle to capture complex three-dimensional plant architectures and angular relationships inherent in canopy structures.

This study investigates two distinct 3D reconstruction approaches for plant trait extraction using a mustard dataset collected through collaboration between Jawaharlal Nehru University and the Indian Agricultural Research Institute (IRAI), Pusa Road, New Delhi. We implement contour-based computer vision with traditional multi-view pose estimation from calibrated images (0°, 120°, 240°, top-view) using 2D keypoint detection, triangulation, and geometric overlay visualization. The second approach employs DUST3R's transformer-based pointmap regression for uncalibrated reconstruction scenarios, eliminating explicit correspondence matching. Both methods are integrated within a unified Streamlit GUI enabling real-time processing of colored PLY point clouds.

Our methodology employs comprehensive phenotypic evaluation including plant height, canopy volume (voxel occupancy and convex hull methods), individual leaf area estimation, stem diameter measurement, and leaf angular orientation analysis. The framework implements robust HSV-based vegetation segmentation with DBSCAN clustering, addressing greenish-yellow flowering stage detection challenges.

This work contributes systematic insights into combining classical contour-based geometry with deep learning for agricultural applications and demonstrates multi-approach validation importance for geometric measurements. The findings have implications for computational plant breeding and stress phenotyping research.

Machine Learning Enabled Examination Feedback System

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The project proposes an Examination Feedback System utilizing machine learning to automate the evaluation of subjective answers in educational assessments. The system will analyse student responses and classify them under a number of predefined topics while making sure that the feedback it generates is tailored and specific to the needs of the students. The primary goal is to ensure improved outcomes through enhanced information to students, where improvement is needed, while helping educators identify performance trends. The system is powered by DistilBERT, which is a transformer-based natural language processing model which is known for its efficiency and contextual knowledge. The proposed system differs from previous manual grading systems in that it offers data-driven information while addressing confidentiality issues with a computationally efficient design. Thus, the examination process becomes scalable, accurate, and data-driven in satisfying the demands of upstream education systems with very limited computational infrastructure.

Bayesian Optimized Gradient Boosting Models for Respiratory Signal Detection using Harmonic and Percussive Source Separation

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The identification, diagnosis and continuous monitoring of abnormal breathing patterns can be enabled through machine learning based respiratory signal detection. This approach introduces a non-invasive way to identify respiratory irregularities from the signal's frequency patterns. However, overlapping audio patterns, instrumental and environmental noises and class imbalance compromises the model's reliability. These limitations can be avoided by using Harmonic Percussive Source Separation (HPSS) as a preprocessing technique, which separates the tonal and transient components of the audio, enhancing the feature extraction while reducing noise in the audio signal. Signal characteristics are then captured by directly extracting features from all three components (signal, harmonic and percussive) which are compiled into comprehensive sets of 237 statistical, spectral, cepstral, chroma, and wavelet features. The underrepresented classes are then balanced using augmentation techniques such as Synthetic Minority Oversampling Technique (SMOTE) and Borderline SMOTE which improves the model's accuracy and generalization. The extracted features are then used to train XGBoost and LightGBM (LGBM) models along with Bayesian Optimization to come up with an optimal set of hyperparameters. Comparing the LGBM models using Borderline SMOTE augmented features with harmonic components of the signals, the unoptimized model reached an accuracy of 90.67% in contrast with the 92.22% accuracy of the model optimized using Bayesian Optimization. Likewise, the optimized SMOTE XGBoost harmonic model outperformed its unoptimized counterpart, achieving 92.35% accuracy over 86.67% accuracy of the unoptimized model. Models using Borderline SMOTE augmentation outperformed standard SMOTE, as seen in the unoptimized LGBM harmonic results, 90.67% for Borderline SMOTE against 87.33% for standard SMOTE. The accuracy of models using the proposed method of spectral decomposition, augmentation using Borderline SMOTE and Bayesian Optimization are consistently better than models without augmentation or optimization, such as 88.17% accuracy for standard LGBM and 85.5% for XGBoost model. This study conclusively shows the practical use of spectral decomposition and Bayesian hyperparameter optimization in developing accurate and reliable gradient boosting classifiers for audio based biomedical signal analysis.

ForenSight: Multimodal Deep Learning for Joint Stamp and Signature Forgery Detection With Pixel-Level Localisation

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Document forgery, especially when it involves stamps and signatures on government and official documents, is a big problem when it comes to verifying the authenticity of documents and stopping fraud. This paper presents ForenSight, a new deep learning system that can detect fake stamps and signatures and locate the exact areas where they were altered, down to the pixel level. The system uses a combination of convolutional neural networks (CNNs) and advanced object detection models like YOLO to extract and analyse both the overall and detailed features of stamp textures and handwriting. By bringing together different visual elements such as texture, shape, and how the signature looks in relation to the rest of the document, the system can catch small signs of fake documents and partial changes that other methods might miss. ForenSight's ability to pinpoint exact areas of manipulation helps forensic experts closely examine and confirm the authenticity of documents. Tests on various datasets show that this model is better than existing methods that look at only one or two types of features in terms of how accurately it detects fakes, how precisely it finds the altered areas, and how fast it works. The system's design allows it to be used in real-time in places like government offices, courts, and banks, offering a strong, easy-to-understand, and all-around solution to tackle difficult cases of document forgery.

Comparative Analysis of Graph Neural Network Architectures for Drug-Drug Interaction Prediction

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Drug-drug interactions (DDIs) represent a critical challenge in modern healthcare, causing adverse effects and reduced efficacy in patients undergoing polypharmacy. With millions of potential drug combinations and limited laboratory testing capacity, computational prediction methods have become essential for patient safety. Traditional machine learning approaches struggle to capture the complex, non-Euclidean relationships inherent in drug interaction networks.

This study investigates Graph Neural Networks (GNNs) for DDI prediction, formulating the problem as link prediction on drug interaction graphs. We implement and compare three representative GNN architectures: Graph Convolutional Networks (GCN) for baseline performance, Graph Attention Networks (GAT) incorporating adaptive attention mechanisms, and GraphSAGE utilizing neighborhood sampling strategies. All models are evaluated on the OGBL-DDI benchmark dataset containing 4,267 drugs and over 1 million documented interactions.

Our methodology employs a dual evaluation framework using both classification metrics (accuracy, precision, recall, F1-score) and ranking metrics (Hits@20, Hits@50). This approach addresses a critical gap in DDI prediction research: the accuracy-ranking paradox, where models achieve high classification scores but fail to rank true interactions appropriately, a crucial requirement for clinical decision support systems.

This work contributes systematic insights into GNN architecture selection for biomedical link prediction tasks and demonstrates the importance of task-appropriate evaluation metrics beyond standard classification accuracy. The findings have implications for computational drug safety and pharmaceutical research.

Defect Detection in Metals and Additive Manufacturing

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The growing usage of modern manufacturing technologies, such as metal forming and metal-based 3D printing, has increased the demand for accurate defect identification, particularly in alloys used in aerospace, automotive, healthcare, and energy applications. Metal alloys are designed to improve mechanical performance, corrosion resistance, and thermal stability. However, they are extremely prone to manufacturing processes such as temperature gradients, solidification rates, phase transitions, oxidation, residual stress, and compositional imbalance. These effects cause significant defects such as porosity, microcracks, inclusions, delamination, partial fusion, and surface contamination. Over time, these flaws can spread under stress, reducing fatigue resistance, and causing catastrophic failure in safety-critical systems. In metal additive manufacturing (3D printing), the layer-wise build process introduces further challenges such as melt pool instability, balling effect, spatter, and uneven microstructures, especially when processing high-performance alloys. Even nanometre scale defects can disrupt mechanical properties, alter grain boundaries, or cause warping an issue particularly crucial for lightweight alloy components used in aircraft, spacecraft, and biomedical implants. Post-processing inspection often reveals flaws too late, making early and real-time defect detection essential to avoid costly reprinting and material waste. Traditional non-destructive inspection technologies like as optical inspection, ultrasonic testing, and rule-based imaging are limited by subjective interpretation, manual reliance, and inadequate scalability for complicated geometries or multi-material alloy systems. Intelligent defect identification is now possible because of advancements in machine vision, deep learning, and high-resolution imaging. Convolutional networks, transformer-based vision models, and no-reference assessment approaches are paving the way autonomous quality control systems discover flaws without established templates. This paper reviews the utilization of these cutting-edge AI methodologies for real-time defect detection across metal manufacturing processes, paves the way for smarter, faster, and more reliable defect detection pipelines that ensure structural safety, reduce production failures, and support the growth of sustainable industrial manufacturing in the era of Industry 4.0.

Key words: Metal defect detection, Additive manufacturing, Machine vision, Deep learning, Non destructive testing, Computer vision.

Balancing Edge-Based Latency and Cloud Flexibility in a Voice-Controlled IoT Vehicle

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The development of voice-controlled robotic vehicles using IoT platforms like the NodeMCU has bifurcated into two dominant architectural paradigms: cloud- dependent online systems and edge-based offline systems. Cloud architectures, which leverage services like Google Assistant, IFTTT, or MQTT, offer extensive natural language flexibility but are critically impaired by high latency, a dependency on stable internet connectivity, and potential privacy concerns. Conversely, edge-based systems using on-device machine learning (e.g., TensorFlow Lite on an ESP32) achieve low latency and privacy but are restricted to a small, rigid, and non-conversational command vocabulary. A significant gap persists in the literature, as most projects fail to address this trade-off or provide a quantitative analysis of system robustness in realistic, noisy environments. This project proposes the design and implementation of a novel hybrid voice control architecture that combines the strengths of both. Our system will utilize a local, edge-based module for instantaneous processing of critical, time-sensitive commands (e.g., "stop," "avoid") while dynamically invoking a cloud- based AI for more complex, conversational instructions. By developing this dual-pathway system, this project aims to create a robust, responsive, and practical vehicle that balances low-latency control with high-level natural language understanding, providing a superior solution to the limitations of existing models.

Keywords—Hybrid Voice Control, Internet of Things (IoT), NodeMCU, ESP32, TensorFlow Lite, Edge Computing, Latency.

IoT-Based Soil Moisture Detection and Automated Seed Plantation Device

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Agriculture continues to face challenges like unpredictable climate patterns, water scarcity, and rising labor costs. To address these issues, our project presents an IoT-based Soil Moisture Detection and Automated Seed Plantation Device designed to make precision farming affordable and accessible for small-scale farmers. This system intelligently monitors soil moisture and temperature and plants seeds only under optimal conditions, ensuring efficient resource utilization and uniform crop germination. Built around the NodeMCU ESP8266 microcontroller, the device integrates soil moisture and temperature sensors, DC motors for mobility, and a servo-based seed placement mechanism for precision planting. The system demonstrated 92% accuracy in detecting suitable planting conditions, maintained seed depth uniformity within 5mm, and achieved 30–40% water conservation compared to manual methods. Costing only INR 4,000, the prototype provides a low-cost alternative to expensive commercial planters, making smart farming feasible for rural and educational applications. With its built-in wireless connectivity, the model supports future IoT expansions such as cloud data logging, mobile monitoring, and AI-based adaptive control. The innovation not only contributes to sustainable farming and resource conservation but also aligns with global Sustainable Development Goals (SDG 2, 6, 9, and 12) by promoting food security, clean water usage, and innovation in agriculture. In conclusion, this project bridges the technological gap between traditional manual methods and modern precision agriculture. It stands as a practical, scalable, and sustainable solution that empowers farmers, supports agricultural education, and showcases how IoT and automation can transform the future of farming.

AI Based Segmentation of Steatosis and Normal Tissues from Liver Biopsy Images

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Accurate quantification of hepatic steatosis from liver biopsy images plays a vital role in diagnosing and managing chronic liver diseases such as non-alcoholic fatty liver disease (NAFLD). Conventional histopathological assessment is labor-intensive, subjective, and susceptible to inter-observer variability. To address these limitations, we present an AI-driven segmentation framework for differentiating steatotic and normal tissue regions in digitized liver biopsy images using deep learning techniques.

The dataset employed in this study was sourced from Mendeley Data, containing annotated liver biopsy images. Comprehensive data augmentation was performed using the Albumentations library to improve model generalization and robustness. The segmentation task was modeled using the DeepLabV3+ architecture, leveraging its encoder-decoder structure and atrous spatial pyramid pooling (ASPP) to capture multi-scale contextual information. Multiple models were trained and evaluated based on metrics such as accuracy, Dice coefficient, and Intersection over Union (IoU).

Preliminary experiments demonstrated that DeepLabV3+ effectively delineates fine-grained tissue structures and achieves promising segmentation performance on validation data. Ongoing work focuses on integrating contrastive learning approaches to enhance feature separability between steatotic and normal tissues, alongside optimization techniques for further performance gains.

The proposed framework highlights the capability of deep learning to automate histopathological tissue segmentation with high precision. It offers potential applications in computer-aided diagnosis, digital pathology, and quantitative liver disease assessment, paving the way for more objective and reproducible diagnostic support in clinical practice.

Study of directional couplers and their applications in microwave engineering

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In this paper, the study focuses on directional coupler - the essential parts of microwave engineering because they are passive, four-port devices that can precisely and minimally sample, divide, or combine power between transmission lines. By allowing a portion of the energy from the input signal to be linked into an auxiliary waveguide, these couplers allow the remaining power to continue unaltered over the main transmission line. Coupling factor, directivity, isolation, and insertion loss are important characteristics of their performance that are crucial in assessing their effectiveness and appropriateness for particular uses.

Directional couplers are essential in real-world systems for a number of functions, including power monitoring, signal sampling, and measuring reflection coefficients in high-frequency circuits and networks. They facilitate crucial measurements like VSWR (Voltage Standing Wave Ratio) and guarantee network stability by enabling precise detection of forward and reflected powers. Modern couplers' excellent accuracy and isolation make it possible to analyze signals simultaneously in both forward and reverse directions, which is essential for accurate load matching and fault investigation in reflectometer systems.

The paper also explores the real-world uses of directional couplers in contemporary radar, communication, and measurement systems, emphasizing their critical function in fault isolation, power monitoring, and signal routing. Recent developments have made it possible to seamlessly integrate couplers into small microwave circuitry by increasing their operational frequency range and miniaturizing their designs. This study aims to give a thorough review of directional couplers, highlighting their significance and adaptability in microwave engineering. It also discusses current issues and new approaches to their design and use.

Index Terms— Transmission line , VSWR.

Attention-Gated U-Net for Semantic Segmentation of Brain Tumors with XAI-Driven Interpretability

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Accurate segmentation of brain tumors from medical images is vital for early diagnosis and treatment planning. This project presents an enhanced **Attention-Gated U-Net (AG-U-Net)** integrated with **Explainable AI (XAI)** to perform precise tumor segmentation while providing insight into the model's decision-making process. Using a publicly available **Brain Tumor Image Dataset** (Roboflow COCO-Segmentation, 2146 images), polygon-based annotations are converted into binary masks to create a high-quality supervised training pipeline.

The proposed model improves upon the classical U-Net by incorporating **Attention Gates** at each skip connection. These gates allow the decoder to selectively focus on tumor-relevant encoder features, reducing background noise and strengthening feature fusion. The network is trained on **512×512** images using a combination of **BCE + Dice loss**, the **AdamW optimizer**, mixed-precision (AMP) training, and learning-rate scheduling. Robust data augmentation including flips, rotations, contrast changes, and noise, further enhances generalization across diverse tumor appearances.

Training for **20-50 epochs** on Google Colab yields stable convergence, achieving **over 50% Dice accuracy** in early experiments. To improve clinical interpretability, the model integrates **Grad-CAM**, **Integrated Gradients**, and **Occlusion Sensitivity**, producing pixel-level attribution maps that highlight regions influencing the segmentation output. Quantitative XAI evaluation using attribution-mask overlap demonstrates strong agreement between model attention and ground-truth tumor regions.

Comparative results show that the Attention-Gated U-Net consistently outperforms the baseline U-Net, particularly in challenging tumor boundaries and low-contrast regions. Overall, the project delivers a complete workflow from preprocessing to segmentation, visualization, and interpretability analysis, making it suitable as both a research-oriented prototype and an educational tool. This work illustrates the potential of combining **attention mechanisms** with **XAI** to create accurate, transparent, and clinically meaningful medical image segmentation systems.

Ego-Relative Event Recognition for Unstructured Indian Traffic

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Understanding how events unfold on Indian roads requires more than detecting objects or classifying actions. The traffic is dense, unstructured, and full of heterogeneous agents whose interactions with the ego-vehicle are often subtle but safety-critical. In this work, we focus on identifying ego-relative events and inter-agent relations by building a new annotated dataset derived from IDD-X and extending it with a symbolic reasoning structure. Each event in our dataset is represented using a unified token format of the form $\langle Z1 \rightarrow Z2 : A1 \rightarrow A2 : \text{Event} \rangle$, capturing spatial zone transitions, actor roles, and the underlying manoeuvre. Using our custom annotation GUI, we manually labelled fine-grained micro-events (e.g., cut-in, merge, overtake, crossing) as well as risk-oriented major events, and linked them through causal relations to represent how one agent’s motion influences another, especially the ego vehicle.

On top of this dataset, we design a baseline model that follows a simple but effective pipeline. First, a dataloader constructs synchronized clips from front and rear camera views. These clips are processed through a lightweight X3D video backbone to extract compact spatio-temporal features. The features are then projected into tokens and passed through a Transformer-based encoder that aggregates multi-view cues. A decoder operates with a fixed set of learnable queries, each corresponding to a potential event instance. Finally, small prediction heads generate the fields of the symbolic event token, along with a soft ordering score that helps sequence events without requiring interval-level annotations.

The outcome is a unified framework that can predict structured event tokens directly from video and align them to the annotated sequence, enabling downstream reasoning on how interactions evolve. Our dataset and baseline model together establish the foundation for interpretable, ego-centric event understanding tailored to the complexities of Indian traffic.

Machine Learning Approaches for Revealing Genetic Patterns in Species Diversity

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Understanding the genetic compatibility between viruses and their hosts is crucial for predicting spill over risks and viral adaptation. This study leverages machine learning techniques to analyse codon usage biases across a diverse set of species, including plants, mammals, and viruses. By applying dimensionality reduction through principal component analysis, the complex codon bias data was visualized in a two-dimensional space, enabling effective classification of species into their respective kingdoms. Among various classifiers, the K-nearest neighbours model demonstrated superior performance in kingdom identification. Further, a specialized model evaluated the genetic similarity between viruses and potential hosts. The former model revealed that species within the same kingdom tend to share compatible codon usage patterns, indicative of their translational and evolutionary relationships. The later model gives insights that contribute to predicting host-virus interactions and assessing zoonotic and spill over risks, emphasizing the potential of machine learning in viral genomics and epidemiology.

Comparative Analysis of Game Theory and Machine Learning for Flood Risk Prediction in Mumbai

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Introduction and Problem Statement Rapid urbanization and the geography of Mumbai followed by intense rainfall in monsoon, lead to severe flooding. Many studies focus on flood prediction, but they overlook the need for a framework to optimize the decision-making for effective mitigation.

Research Objective This paper proposes a novel comparative analysis of Machine Learning (ML) and Game Theory (GT) to develop an integrated decision-support system for flood risk in Mumbai.

Methodology: Gathering the historical data of Mumbai (20-25years), based on the indicators such as Slope, Aspect, Elevation, TPI, TRI, TWI, Land-Use/ Land cover, Rainfall, Mumbai Boundary, Curvature, Drainage Density and Distance from River. The study is a two-step process Machine Learning Prediction and Game Theory Strategy ¹

Machine Learning Prediction First, we use ML models such as XGBoost, Random Forest & ANN to do flood probability mapping. Then these models will be trained on the data and performance will be evaluated using standard metrics (F1-score, AUC).

Game Theory Strategy This component will implement and compare Game Theory strategies such as Borda count, Condorcet. Decision making can be done backed by this strategy in order to optimize the flood mitigation process.

Expected Implications This study is thus expected to perform a comparative analysis, integrating advanced machine learning forecasting with strategic game-theoretic-based policy models for the flood risk in Mumbai. The expected outcomes are to provide a strong, dual-purpose framework that will not only enhance predictive accuracy but will also provide a data-driven basis for strategic coordination in pursuit of more resilient urban planning and flood management.

Modeling Functional Disruption and Cascading Flood Impacts on Critical Infrastructure in Data-Scarce Cities

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Urban flooding, intensified by climate change, poses a significant and growing threat to critical infrastructure (CI) in data-scarce cities, which are often overlooked in research. Current rapid flood assessment frameworks, while scalable, are constrained by significant limitations. They typically rely on a binary (inundated/not inundated) impact model and use non-physical geometric buffers to estimate peak flood extent from delayed satellite imagery. This approach fails to capture the partial degradation of infrastructure function and the cascading failures that characterise the true impact of a flood. Conversely, advanced functional impact models are too data-intensive for these environments, creating a critical methodological gap.

This paper addresses this gap by developing and validating an enhanced rapid flood impact assessment framework that synergises the speed of remote sensing with the analytical realism of functional and systemic models. Using the July 2023 extreme flood event in Delhi, India, as a case study, our methodology integrates several open-source tools and datasets into a novel end-to-end workflow.

First, a post-event flood extent is mapped using Sentinel-1 Synthetic Aperture Radar (SAR) imagery. This extent, combined with a Digital Elevation Model (DEM), is then processed using the Floodwater Depth Estimation Tool (FwDET), a GIS-based algorithm that calculates water depth from an inundation map and a DEM. This crucial step generates a physically plausible flood depth map, enabling the application of a depth-disruption function to model a realistically degraded transportation network. Finally, a service area analysis is conducted to quantify the loss of public accessibility to critical services, while first-order cascading effects, such as power outages from inundated substations, are traced using network data from OpenStreetMap (OSM). This research delivers a scalable, replicable, and analytically robust methodology to provide urban planners in vulnerable cities with a powerful tool to enhance flood resilience.

Deep Learning-Based Automated Function Prediction for Uncharacterized Proteins using Protein Language Models and Hierarchical Label Propagation

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Accurately mapping biological functions onto protein sequences is one of the most fundamental challenges in the area of bioinformatics, which represents an increasingly essential area in the fields of drug development and cellular mechanisms. The exponential growth of sequencing data from genomics far outpaces experimental validation of functions, thus resulting in an "annotation gap." This work builds on and presents a comprehensive deep learning framework for predictions of GO terms from amino acid sequences developed for CAFA 6.

This work proposes a methodology that uses ESM2, a state-of-the-art transformer-based protein language model, to create high-dimensional embeddings that capture the semantics of protein sequences on both evolutionary and structural levels. These embeddings are then used to train a custom neural network classifier optimized through IA weighted loss to ensure the model focuses on specific, information-rich functional terms over generic annotations. Finally, this work implements a hierarchical label propagation algorithm post-processing step to enforce consistency within the Gene Ontology DAG by ensuring child term predictions are logically supported by their parent terms.

This study demonstrates that transfer learning from pre-trained language models significantly outperforms traditional homology-based methods, particularly for sequences with low sequence similarity to known proteins. The model was then deployed to predict probabilistic functional annotations for more than 224,000 uncharacterized proteins across Molecular Function (MF), Biological Process (BP), and Cellular Component (CC) domains. These results provide a scalable, high-throughput solution to the protein annotation bottleneck, offering reliable candidates for downstream experimental validation in systems biology and medical research.

Keywords: Protein Function Prediction, Gene Ontology, ESM2, Deep Learning, Natural Language Processing, Bioinformatics, Hierarchical Classification.

Index Terms— Computational Biology, Machine Learning, Neural Networks.

Fabrication of a Bubble-Assisted Pneumatic Module for Semi-Submerged Waste Retrieval

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Among the most crucial yet not-discussed environmental concerns is the accumulation of non-biodegradable waste, especially plastics, on the surface of rivers and lakes. The Yamuna, Delhi's lifeline, is choked with 58% of the city's waste, endangering lives and ecosystems, (CPCB,2021). According to the UNEP (United Nations Environment Programme, 2022), over 11 million metric tons of plastic enters water environment globally each year in turn harms biodiversity, contaminates drinking water sources, and affects over 2 billion people who rely on surface water for their daily needs. To tackle this issue, we propose a water surface cleaning bot that operates along the shore, detecting and collecting floating garbage to help clean water bodies. The system comprises of IoT-based sensors, real time image recognition, and GPS based navigation which autonomously detects, collects and categorizes floating waste, transmitting real time data featuring a virtual geo-fencing module to trigger alerts when unauthorized discharge activities are detected. Unlike conventional manual clean-up methods –which are inefficient, labour intensive and risky– our bot uses computer-based algorithms and mobile applications that can operate continuously with minimal human intervention. This hybrid solution not only ensures curative clean-up operations but also enforces preventive surveillance, making it a significant advancement in smart water resources for urban ecosystems.

Keywords – AI- based pollution detection, aquatic robotics, IoT, surface water sanitation, urban water management, waste monitoring, Yamuna River.

An Integrated Approach to Audio Signal Processing: FM Simulation, Noise Reduction, and Speaker Recognition Using Deep Learning

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The concept of audio signal processing is integral to communication, security, and human-machine interaction scenarios where it's necessary for speech clarity and accurate identification of speakers. This work proposes an approach that combines traditional signal-processing methods with recent advances in deep learning in order to create a common pipeline for FM simulation, noise reduction, and speaker recognition. The work begins with the simulation of frequency modulation (FM) to understand how the signal behaves under various channel conditions and understand the distortion effects introduced through transmission.

After demodulation, noise reduction occurs using a multi-stage processing pipeline that employs high-pass filtering, noise gating, dynamic range compression, and Short-Time Fourier Transform (STFT)-based spectral subtraction. At the same time RNNoise, a recurrent neural network based noise suppression, has been included to further improve performance when the noise is non-stationary or a complex signal. This hybrid approach greatly enhances speech intelligibility and reduces interference while keeping the integrity of the voice signal.

For recognizing speakers, audio signals are first transformed into Mel-spectrogram representations that can effectively preserve perceptual frequency information. Then, a custom Convolutional Neural Network (CNN) is trained on the spectrograms to learn to distinguish features unique to each speaker. The system has exhibited high classification performance across evaluation samples, and the confusion matrix reveals reliable separation of the individuals observed, with infrequent misclassifications. The experimental results substantiate that the complete processing pipeline provides strong noise suppression, as well as reliable speaker identification.

More broadly, this work demonstrates the potential of bridging mode analysis, advanced techniques for noise reduction, and deep learning based feature extraction for tackling issues in audio applications of the 'real-world'. The proposed approach provides promising implications for both biometric authentication technologies and intelligent voice interfaces, and in communication systems.

Development of a Smart Molecular Beacon-Based Genosensor for Detection of Human Rotavirus A

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The rotavirus-led fatal infantile gastroenteritis in the globe demands a portable, specific, and low-cost diagnostic tool for its timely detection and effective surveillance in a mass population. Consequently, the design and development of an advanced biosensing technique for its detection is of paramount importance. A highly conserved 23-nucleotide sequence, 5' GCTAGGGATAAGATTGTTGAAGG 3', was identified by a human rotavirus A VP6 gene sequence analysis and designated as the target. A molecular beacon of 33 nucleotides was designed with the sequence 5'[Fluorescein] ATAGTCCTTCAACAATCTTATCCCTAGCACTAT[Dabcyl]3', incorporating stem and loop regions. Secondary and tertiary structure characterizations confirmed the desired stem-loop structure without internal secondary structures. The thermal stability of the molecular beacon-target complex was studied using a temperature vs. Gibbs free energy change plot, melting curve analysis based on absorbance vs. temperature, and an experimental fluorescence resonance energy transfer melting assay. The melting temperature of the molecular beacon-target complex was experimentally determined to be 62 °C. The spectral analysis showed fluorescence restoration in the presence of the synthetic VP6 target. The assay conditions were optimized with an excitation wavelength of 470 nm and a 10-min incubation time. The assay demonstrated a linear correlation between fluorescence intensity restoration and target concentration, with a limit of detection of 18.8 nM. Interference studies with single mismatch, double mismatch, and scrambled targets revealed that the molecular beacon has strong specificity for the VP6 target, effectively discriminating against non-target sequences. This work demonstrates the molecular beacon's potential as a sensitive and specific tool for detecting rotavirus A VP6 gene, with promising applications in diagnostic assays for the rotavirus disease management.

A Machine Learning–Enhanced Framework for Delta Modulation and AWGN Channel Reconstruction

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Delta Modulation (DM) is widely recognized for its structural simplicity and bandwidth efficiency in digital communication systems; however, its performance is significantly affected by inherent drawbacks such as granular noise, slope overload distortion, and vulnerability to channel-induced impairments. These limitations become more prominent when DM signals traverse an Additive White Gaussian Noise (AWGN) channel, where the presence of random noise further degrades signal fidelity and reduces reconstruction accuracy. With recent advancements in computational intelligence, machine learning (ML) provides a promising avenue for enhancing classical communication models. This study proposes a machine-learning-enhanced framework to improve the reconstruction of DM signals transmitted through AWGN environments.

The work begins by implementing a classical DM encoder–decoder system and analyzing its behavior under varying SNR conditions. Distortion characteristics, error patterns, and waveform degradation are examined to establish a performance baseline. To overcome these limitations, an ML-based reconstruction architecture is designed and trained using preprocessed noisy DM datasets, enabling the network to learn optimal mapping functions for accurate and noise-resilient signal recovery. Comprehensive training, validation, and testing procedures are employed to evaluate the model’s performance using time-domain analysis, error measurement, bit-level accuracy, and waveform fidelity.

Experimental results demonstrate that the proposed ML-based framework achieves significantly improved reconstruction quality, yielding a 2–3× reduction in error compared to classical DM decoding. The model exhibits superior noise suppression, enhanced slope approximation, and strong robustness across diverse AWGN conditions.

Overall, this study establishes the effectiveness of integrating machine learning with traditional modulation techniques. The findings highlight the potential of ML-enhanced approaches for next-generation communication systems, offering improved reliability and performance in practical noisy environments.

Omnidirectional Vivaldi UWB MIMO Antenna for Next-Generation Applications

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The integration of Multiple-Input Multiple-Output (MIMO) technology with Ultra-Wideband (UWB) systems is essential for advancing next-generation wireless communications. The Vivaldi antenna is a key component in this field, known for its wide bandwidth, compact design, and suitability for UWB applications. Its characteristic exponential taper profile ensures a smooth impedance transition and optimal radiation characteristics across the operating band. However, designing effective 3D MIMO antennas using these elements faces persistent challenges, including high mutual coupling, difficulty in achieving consistent impedance matching, and ensuring stable radiation patterns across the full UWB spectrum (3.1–10.6 GHz).

This paper presents the design and simulation of a novel Vivaldi UWB MIMO antenna engineered to overcome these limitations. The proposed solution features a four-element array arranged in a circular, ring-shaped configuration with an outer diameter of 260 mm. This dual-layer structure, with 90° angular spacing between elements, is specifically designed to provide uniform omnidirectional coverage while enhancing spatial diversity and minimizing mutual coupling.

Simulations conducted using CST Microwave Studio Suite validate the design's superior performance. The antenna achieves effective impedance matching, with the reflection coefficient (S_{11}) remaining below -10 dB across the entire operational frequency band. Furthermore, the ring configuration provides excellent isolation, with mutual coupling (S_{21}) between adjacent elements maintained below -20 dB. Key MIMO performance metrics confirm the design's effectiveness: the Envelope Correlation Coefficient (ECC) is held below 0.5, and the Diversity Gain (DG) consistently ranges between 9.95 and 10, approaching the theoretical ideal. These results demonstrate that the proposed ring-shaped antenna is a practical and robust solution for advanced wideband communication applications.

SMART DUSTBIN USING NODEMCU(ESP8266)

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The increasing volume of waste generated in urban and semi-urban areas has created significant challenges in maintaining hygiene, ensuring timely waste collection, and promoting environmentally responsible disposal habits. Manual handling of dustbins often leads to direct contact with contaminated surfaces, posing health risks and contributing to unhygienic surroundings. Furthermore, traditional dustbins lack intelligent features such as waste classification and automated operation. To overcome these limitations, this project proposes the design and development of a **Smart Dustbin using NodeMCU (ESP8266)** that automates lid movement, detects nearby objects, and identifies waste as wet or dry.

The primary objective of this smart dustbin is to provide a **contactless, efficient, and user-friendly waste disposal system** that enhances sanitation in both private and public spaces. The system uses an ultrasonic sensor to detect a user's hand or an approaching object, triggering a servo motor to open the lid automatically. Additionally, a raindrop sensor is integrated to differentiate between wet and dry waste, enabling basic waste segregation at the source. The NodeMCU microcontroller processes all sensor inputs and coordinates system behavior with minimal delay and low power consumption.

Experimental results show that the smart dustbin responds quickly to object detection, maintains stable lid operation, and reliably identifies moisture-based waste categories. These features collectively improve hygiene, reduce physical contact, and encourage better waste management practices. The project demonstrates how IoT-enabled embedded systems can enhance everyday sanitation infrastructure. Future improvements may include fill-level monitoring, wireless alerts, solar-powered operation, and advanced AI-based waste classification.

A Study on Anechoic Chamber for Antenna Radiation Pattern Measurement

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In this paper, Antenna radiation pattern is an important factor in understanding how an antenna sends and receives signals in different directions. For RF and microwave systems, the accuracy of this measurement directly affects antenna performance analysis. However, taking these measurements in open areas or normal laboratory spaces becomes difficult because reflections from walls, surrounding objects, and external signals interfere with the readings. Due to this, it becomes necessary to use a controlled environment for measurement.

An anechoic chamber is a specially designed room that absorbs electromagnetic waves and prevents reflections. Because of this property, the chamber behaves like a free-space environment and allows antennas to be tested without unwanted interference. In this study, the complete working of an anechoic chamber is explored, including its structure, internal layout, and the type of absorbing materials used to reduce reflections. The project also discusses how antennas are positioned, the general testing procedure, and how the chamber helps in obtaining stable measurements of directional properties such as gain, beamwidth, side lobes, and polarization.

This project helped us understand why an anechoic chamber is considered a standard requirement for antenna measurements in the RF and microwave field. It also provided a clearer idea of how academic and research labs use controlled environments to maintain precision during testing.

Index Terms— Anechoic Chamber, Antenna Radiation, Pattern Measurement.

Designing a Smarter Communication System using an End-to-End Deep Learning Autoencoder

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The design of physical layer (PHY) communication systems has traditionally relied on a block-based approach, where components like channel coding and modulation are designed and optimized independently. While effective, this modular sub-optimization is not guaranteed to be globally optimal. This paper investigates an end-to-end (E2E) deep learning approach, modeling the entire transmitter-receiver chain as a single autoencoder. This allows the system to be jointly optimized through gradient descent to discover novel, high-performance communication schemes.

We implement an autoencoder in Python using TensorFlow/Keras and the NVIDIA Sionna library. The transmitter (encoder) and receiver (decoder) are modeled as multi-layer perceptrons (MLPs). The system is designed to transmit $k = 4$ bits per message, mapping them to $n = 2$ complex symbols, corresponding to a spectral efficiency of $R = 2.0$ bits/symbol. The model is trained through a differentiable Additive White Gaussian Noise (AWGN) channel layer, with a fixed training signal-to-noise ratio (E_b/N_0) of 7 dB. We evaluate the system's performance by simulating the Bit Error Rate (BER) over a range of E_b/N_0 values from 0 to 10 dB. The results are benchmarked against a traditional 16-QAM system, which is a human-designed standard for $k = 4$ bits. Our E2E autoencoder significantly outperforms the 16-QAM baseline across the entire SNR range. At an E_b/N_0 of 8 dB, the learned system achieves a BER of approximately 1.4×10^{-6} , demonstrating a coding gain of over 2 dB compared to the standard scheme, which only achieves a BER of 4.4×10^{-3} .

To understand this performance gain, we visualized the constellation learned by the transmitter. Unlike the rigid 4×4 square grid of 16-QAM, the autoencoder discovered a novel, non-square, quasi-circular constellation. This learned geometry maximizes the average Euclidean distance between constellation points, making it inherently more robust to noise. This work successfully demonstrates that an autoencoder can, without prior human knowledge, learn a complete high-performance communication system from scratch. This E2E approach shows significant promise for designing future AI defined air interfaces and adaptive communication systems.

Performance analysis of 6T SRAM cell using CMOS 90nm technology

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The importance of effective SRAM design has grown due to the increasing requirement for high performance and low-power memory systems in contemporary VLSI circuits. This research presents the performance analysis of 6T SRAM cell at 90nm CMOS technology. Cross-coupled inverters and access transistors are used in the cell architecture to ensure data retention, and stability is assessed at different supply voltages using Cadence Virtuoso, with simulation results are verified with Spectre. The results show that lower supply voltage reduces dynamic power dissipation, which affects speed and stability because of sensitivity to leakage and noise. Average power consumption 3.6 μ W, the 6T SRAM cell is appropriate for embedded memory applications lowers energy consumption. The 6T SRAM cell comprises two cross coupled CMOS inverter stores 1 bit value and their complement, helps to check how the cell behaves when the cell behaves when the voltage (VDD) changes. We studied what happens during read and write operations, how much leakage current flows, and how much total power is used. But at the same time, the circuit becomes slower and less stable. So we need to choose a voltage that keeps the circuit both fast and stable. This study lays the groundwork for future optimization in sophisticated memory architectures by highlighting the trade-offs between power, performance, and reliability. The paper demonstrates that using 90nm CMOS technology, the 6T SRAM cell achieves excellent energy efficiency without losing much performance. The results will help VLSI engineers and researchers design next-generation memory circuits that are smaller, faster, and more reliable.

Keywords— 6T SRAM, CMOS 90nm Technology, Power Consumption, Leakage Current, Stability Analysis, Cadence Virtuoso, Low-Power VLSI Design

Mamba-UNet: An Efficient State Space Model Architecture for Medical Image Segmentation

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Medical image segmentation is fundamental to computer-aided diagnosis and clinical decision support, particularly in endoscopic and radiological workflows. While convolutional neural networks and transformer-based architectures have achieved strong performance, their ability to capture long-range dependencies often incurs high computational and memory overhead. This paper introduces Mamba UNet, an efficient hybrid framework that incorporates the Mamba Selective State Space Model (SSM) into the UNet encoder-decoder structure. By replacing the bottleneck layers with Mamba blocks, the proposed architecture enables global context modeling with significantly reduced computational cost. Experiments conducted on the Kvasir-SEG polyp segmentation dataset demonstrate that Mamba-UNet achieves competitive Dice and mIoU performance compared to state-of-the-art UNet and transformer baselines. Ablation studies confirm the contribution of Mamba layers to improved boundary preservation and robustness to small structures. Cross-dataset evaluation on CVC-ClinicDB indicates strong generalization capability.

Optimized Microstrip Antenna Design for High-Bandwidth Wireless Communication and Sustainable IoT Systems

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This paper presents a comprehensive comparative analysis of a Proximity-Coupled Microstrip Patch Antenna (MSA) designed for operation in the highly utilized 2.4 GHz Industrial, Scientific, and Medical (ISM) band, crucial for modern IoT and Wireless Communication systems. The study investigates the performance impact of varying both the antenna's feed structure and the substrate material. Two primary feedline configurations—a simple microstrip feedline and an innovative fork shaped feedline—are compared across four diverse substrates: PTFE (Teflon), Polyimide, Duroid 5880, and Roger RO 3035. Key findings demonstrate a significant trade-off and optimization capability: On the Duroid 5880 substrate, the fork- shaped feedline achieved a desirable 50 MHz Bandwidth, compared to the 30 MHz bandwidth of the simple feedline, though the simple feedline maintained a superior reflection coefficient (S_{11}) of -32 dB. Crucially, the Polyimide substrate, known for its flexibility and suitability for compact/wearable electronics, showed that the fork-shaped feedline could increase the bandwidth while maintaining a highly efficient match ($S_{11} < -20$ dB). Across all substrates, the fork-shaped design consistently demonstrated the potential to enhance Directivity and/or Bandwidth by providing an additional tuning element for impedance matching and current distribution control as evidenced by

Across all substrates, the fork-shaped design consistently demonstrated the potential to enhance Directivity and/or Bandwidth by providing an additional tuning element for impedance matching and current distribution control, as evidenced by improved gain patterns simulated in CST Studio Suite.

Attention U Net based Segmentation of Liver and Liver Mass from B-Mode Ultrasound Images of Fatty Liver

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Medical ultrasound is a primary diagnostic modality for hepatic diseases due to its cost-effectiveness and non-invasive nature. However, the manual segmentation of liver organs and lesions is hindered by the inherent presence of speckle noise, low contrast, and shadowing artifacts. This paper presents a deep learning framework for the automated, multi-class segmentation of the liver and hepatic masses. We propose an Attention U-Net architecture, which integrates attention gates into the skip connections of a standard U-Net. This mechanism suppresses irrelevant regions in the feature map while highlighting salient features relevant to the target structures.

The model was trained on a dataset of annotated abdominal ultrasound images using robust data augmentation techniques to improve generalization. To address class imbalance between the background, liver, and lesions, we employed a compound loss function combining Cross-Entropy Loss and Dice Loss. Experimental results on the test set demonstrate the efficacy of the proposed approach, achieving an Average Foreground Dice Score of 0.7286. Specifically, the model achieved a Dice Score of 0.8340 for Liver segmentation and 0.6231 for Mass detection. These results suggest that the Attention U-Net is a viable tool for assisting radiologists in the rapid and accurate diagnosis of liver pathologies.

Ground Vibration Monitoring Using Non-Contact Sensor Systems

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Research entitled "The Design and Development of a Prediction Model for Ground Vibrations Using Sensor-Based Network" seeks to forecast ground vibrations in many different environments using sensor technology. The environment, human comfort, and surrounding structures can all be highly affected by ground vibrations caused by mining, construction, and transportation. For risk calculation, infrastructure design, and mitigation methods, a reliable and accurate prediction model is very valuable. The initial work in this research is to establish a network of sensor nodes that are placed strategically across areas that are prone to ground vibrations. These sensor nodes capture and transmit real-time readings to a central processing unit, such as acceleration, displacement, and frequency content. The prediction model is trained and verified from the data collected by sensor nodes in the network. Machine learning algorithms such as support vector machines, random forests, and neural networks can be utilized to handle large datasets and establish credible prediction models. The efficiency of the model and the generalisation will be enhanced by employing a feature-rich method to derive significant parameters and decrease data dimensionality. Once the prediction model is established, it is thoroughly tested against field measurements to ensure accuracy and reliability. Comparisons to be established with industry benchmarks and current empirical models to evaluate the performance of the model. Moreover, the sensitivity of the model to soil type, construction work, and weather conditions is examined to better understand its applicability. A novel approach to ground vibration prediction, leveraging the power of sensor-based networks and advanced machine learning techniques is presented through the research. The implications of the model also extend to a wide range of applications such as the assessment of structural integrity, environmental impact assessment and urban planning. Continuous data collection and model refinement has the potential for enhancing the prediction model. Recommendations for data transmission, optimizing sensor placement and model training are provided to improve the system's cost effectiveness.

Digital Twin-driven Optimization framework for Additive Manufacturing Systems: Real Time simulation, Defect Prediction - A Comprehensive review

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Additive Manufacturing (AM) has become a central technology within Industry 4.0, offering unmatched geometric flexibility, rapid prototyping, and material-efficient production for aerospace, automotive, biomedical, and tooling sectors. However, despite these advantages, AM continues to face persistent limitations such as thermal gradients, residual stresses, porosity formation, surface irregularities, and build-to-build variability. These issues not only affect printability but also disrupt the crucial relationship between process parameters, microstructure evolution, and the final mechanical performance of the part. Several approaches such as process parameter tuning, in-situ sensing, topology optimization, support-structure refinement, and thermal–mechanical simulations have been explored to mitigate these issues. While these methods improve specific aspects of the process, they are often reactive, limited in scope, and unable to adapt continuously during fabrication.

Considering these developments and the need for more reliable AM processes, this study explores the use of a Digital Twin (DT) framework for metal and polymer additive manufacturing. A DT integrates real-time sensor data acquisition, multi-physics simulations, and predictive machine learning models to construct a continuously updated virtual representation of the fabrication process. By capturing melt-pool dynamics, heat flow behavior, laser–material interaction, deposition patterns, and phase transformations, the DT can identify defect formation pathways and correlate them with printing conditions. Existing state-of-the-art studies demonstrate how DTs improve process transparency, enable parameter optimization, and support in-situ adaptive control for enhanced dimensional accuracy and mechanical reliability.

This review highlights progress made in DT-enabled AM, identifies current limitations such as model scalability, computational cost, and sensor integration challenges, and outlines future research opportunities. These include autonomous self-correcting systems, hybrid physics–ML predictive models, standardized DT architectures for AM machines, and their integration into smart manufacturing lines to achieve fully intelligent, self-optimizing AM ecosystems.

Keywords: Additive Manufacturing, Digital Twin, Physics informed Machine Learning and Smart Manufacturing

Characterization Study of Brass Gun Cartridges coated with Self-Healing Polymers using SEM

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Brass, an alloy of copper, is commonly used for the fabrication of gun cartridges. It is also a practical and well-established choice due to its excellent corrosion resistance, reliable sealing in firearm chambers, ease of manufacturing, and good reloading capabilities. Its fabrication processes are widely standardized, making it cost-effective and accessible. This material will provide a solid balance of durability, performance, and manufacturability suited for most general ammunition applications. DGEBA (Diglycidyl Ether of Bisphenol A), epoxy as a self-healing polymer when coated, provides an excellent balance of mechanical strength, adhesion with good toughness and corrosion resistance. This polymer responds to damage stimuli by autonomously repairing microcracks and minor surface damage, thus extending the service life and reliability of the cartridges. The combination of brass's innate properties with the smart functionality of self-healing epoxies yields a competitive edge over conventional uncoated brass cartridges. Scanning Electron Microscope (SEM) was used to determine the microstructure and surface morphology, and composition. As these fired cartridges are subject to dynamic environments, cold working pressures come into play. The characterization techniques carried out gave detailed insights. Special emphasis is laid on the study of fracture areas, grain boundaries, and its effect on the use of material. With the addition of self-healing polymers, we are increasing the workability of gun cartridges at different conditions. This integrated approach supports the development of robust ammunition suited to diverse operational conditions and fulfills requirements across military and civilian applications, making DGEBA-coated brass cartridges a promising advancement in ammunition technology. It deals with both military graded and civilian graded gun cartridges. There is much advancement going on to combine with advanced materials to maximize performance with better economic aspects. And various other characterization techniques are suggested for detailed analysis of the material.

Design And Analysis of Composite Leaf Springs For Enhanced Vehicle Efficiency

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In today's world, where climate change and environmental sustainability are major concerns, the automobile industry is working hard to make vehicles lighter and more energy-efficient. One important part of a vehicle that can help achieve this is the **leaf spring**—a key component in the suspension system that supports the vehicle's weight and absorbs shocks. Traditionally, leaf springs are made from heavy steel, which adds to the overall weight of the vehicle and increases fuel consumption.

This study explores the use of advanced composite materials, such as carbon/epoxy and glass/epoxy or some other combination, to replace conventional steel leaf springs. Using computer-aided design tools like SOLIDWORKS and simulation software like ANSYS, researchers created and tested different types of leaf springs under various load conditions. The results showed that composite leaf springs are not only much lighter—reducing weight by up to 70%—but also perform better in terms of stress resistance and flexibility. A hybrid version, combining steel and glass epoxy, also showed promising results with a 34% weight reduction and significantly lower stress levels.

By switching to composite materials, vehicles can become lighter, which means they use less fuel for propelling and emit fewer greenhouse gases. This directly supports global efforts to reduce carbon footprints and combat climate change. Lighter vehicles also require less energy to manufacture and transport, further reducing environmental impact.

Moreover, composite materials are corrosion-resistant and have a longer lifespan, which means fewer replacements and less waste. Their high strength-to-weight ratio makes them ideal for electric vehicles, which benefit greatly from reduced weight to improve battery efficiency and driving range.

In conclusion, replacing traditional steel leaf springs with composite alternatives is a smart move for both vehicle performance and environmental sustainability. It's a step forward in designing greener, more efficient automobiles that align with the goals of reducing pollution and preserving our planet for future generations.

Keywords-Composite Materials, Leaf Spring, Weight Reduction, Finite Element Analysis, Sustainable Automotive Design.

Sustainable Biocomposites for Advanced Manufacturing: A Comprehensive Review of Processing, Printability, and Functional Applications

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Biocomposites are the engineered materials derived from natural fibers, biopolymers, and bio based reinforcements which have emerged as sustainable alternatives to conventional synthetic composites due to their low environmental impact, biodegradability, and favorable mechanical performance. Recent advances in processing technologies have significantly enhanced the manufacturability, functional tunability, and design freedom of biocomposites, enabling their adoption across automotive, biomedical, packaging, construction, and aerospace sectors. This review provides a comprehensive assessment of biocomposite materials, focusing on fiber extraction, matrix selection, interface engineering, and fabrication strategies that determine their performance and scalability. Special emphasis is placed on additive manufacturing, particularly 3D and emerging 4D printing techniques, which have opened new avenues for complex geometries, adaptive structures, and stimuli-responsive biocomposites. The integration of natural fibers and bio-polymers into printable feedstocks, process–structure–property relationships during extrusion or vat-based printing, and challenges such as printability, anisotropy, moisture sensitivity, and long-term durability are critically analyzed. Furthermore, the review highlights current innovations in smart biocomposites enabled by 4D printing, where time-dependent shape transformation and multifunctionality allow applications in soft robotics, tissue engineering, flexible electronics, and sustainable product design. Future research directions are outlined, emphasizing material formulation, hybrid bio-reinforcement strategies, process optimization, and digital manufacturing approaches to fully realize the potential of biocomposites in next-generation sustainable engineering systems.

Keywords: Biocomposites; Natural Fibers; Biopolymers; Sustainable Materials; Manufacturability; Additive Manufacturing; 3D Printing; 4D Printing; Smart Materials; Stimuli Responsive Composites; Bio-based Polymers; Green Manufacturing.

Deepfakes: Creation, Detection & Mitigation: Understanding AI-Generated Synthetic Media Technologies

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Deepfakes are artificial media created with the help of high-quality AI models and are a serious threat to the digital credibility, stability, and the security of society. This is a comprehensive poster on deepfake creation algorithms, such as autoencoders, GAN models, and diffusion-based generative models. Based on the latest academic research, it explores state of the art detection methods including convolutional and transformer-based forensic models, physiological signal detection, frequency-domain artifact detection, and multimodal detection methods. Furthermore, it identifies proactive mitigation affairs that include watermarking, blockchain-provenance, policy-based controls. Through the integration of empirical evidence on media forensics and AI security studies, the poster would provide the research community, practitioners, and policymakers with a unified view of the dynamic threats and novel defence strategies to deal with deepfakes.

Petri Net and Graph Theory–Based Modeling of Air Pollution Dynamics in Delhi

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Air pollution in Delhi remains a significant environmental and public health concern, particularly during winter when stagnant atmospheric conditions and increased emission activities sharply elevate pollutant concentrations. To capture the complex interactions among major pollutants such as PM_{2.5}, PM₁₀, NO₂, O₃, CO, SO₂, and NH₃, this study presents an integrated Petri net and graph theory–based modeling framework that treats Delhi’s air pollution as a dynamic, interconnected system. In the Petri net model, places represent pollutant states and emission sources, transitions denote chemical reactions, meteorological changes, and emission events, and tokens correspond to the quantities of pollutants. The system’s incidence matrix and reachability graph are used to analyze how transitions drive pollutant evolution over time, while invariant analysis identifies stable relationships and recurring emission pathways contributing to persistent poor air quality. Complementing this, a graph-theoretic network is constructed where nodes represent pollutants and emission sources, and edges capture influence, transformation, or interaction pathways. Graph metrics such as degree and betweenness centrality, clustering coefficients, and shortest paths reveal dominant pollutants, critical emission nodes, and high impact pathways that drive rapid AQI deterioration. Using air quality datasets, the integrated model uncovers feedback loops, transition-driven pollution surges, and structural vulnerabilities in the pollution network, offering a robust visual, analytical, and predictive tool to support policymakers and environmental agencies in designing data-driven, system-level strategies for effective air quality management in Delhi.

Keywords: Petri nets, Graph theory, Air pollution modeling, Delhi air quality, Dynamic systems, Pollution network analysis, Reachability graph, Centrality metrics, Invariant analysis, Environmental modeling.

AI and Remote Sensing Based Yield Estimation and Crop Classification

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Accurate crop mapping and yield estimation are critical for enhancing agricultural productivity and resource management. This study presents an advanced framework for generating precise wheat crop masks and predicting yield in the Karnal district of Haryana, India, using multi-temporal Sentinel-2 satellite data and Artificial Intelligence techniques. The crop mapping module integrates supervised (Maximum Likelihood Classification) and unsupervised (ISODATA) methods with NDVI time-series profiles to address spatial heterogeneity and improve crop area delineation. Building on this foundation, six machine learning models were evaluated for yield prediction using NDVI and NDWI time-series data along with village-level yield records. Among these, Random Forest Regression achieved the highest coefficient of determination ($R^2 = 0.84$), followed closely by Gradient Boosting Regression ($R^2 = 0.81$), demonstrating strong predictive accuracy. The integration of remote sensing, IoT, and advanced AI-driven analytics highlights the potential for scalable, automated systems in precision agriculture, supporting sustainable farming practices, biomass estimation, and food security initiatives.

Real-time human activity recognition using smartphone sensors using federated learning

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Recently, the issues in recognizing human activity have been well addressed in various research articles. The research challenge appears whenever the smartphone starts to recognize human activity in a real-time environment. There are different application area that covers this situation, such as yoga positions, activity positions, and more. To extract sensor data in real-time, we have started using the WebSocket technique. Moreover, the accelerometer and gyroscope are two essential sensors, typically found in smartphones, that are responsible for providing activity sensor data. To implement this proposed technique, we need to address the following issues:

- Presence of continuity while streaming of activity sensor data
- Finding the sampling rate consistency
- Sensor employment in smartphone issue (X, Y, Z axis)
- Smartphone placement orientation
- With respect to the activity type that challenges the placement of the smartphone sensors
- Most importantly, it captures the computation capability of the sensing device.

The listed points are presented as obstacles that need to be addressed on the way to implementation. Initially, we are working to develop a small and powerful computing device that makes sensor data available, similar to the Raspberry Pi and others.

Instead of the classical learning model, we focus on using a deep recurrent learning model. This model can handle sequential streaming data and memorization to identify learning patterns in activity. To improve the performance of the recurrent model, we are utilizing the concept of the federated learning technique. This is a decentralized learning technique that learn from various local data points. Once the model is trained on the regional level, it is sent back to the centralized level for aggregation purposes. Furthermore, the trained model started to recognize the human activity patterns from streaming sensor data.

Computational Modelling of Hamiltonian Cycles in Graphs using Python

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This work investigates Hamiltonian cycles through computational modelling to analyze how a graph can be traversed by visiting each vertex exactly once. Because identifying Hamiltonian cycles is an NP-complete problem, it remains computationally challenging despite its relevance in practical areas such as routing, scheduling, and network design. Using Python and the NetworkX library, we generated random graphs, grid graphs, and custom connected graphs to test for Hamiltonian cycles with a backtracking-based algorithm. The algorithm performed well on smaller graphs but slowed significantly as graph size increased, and our results showed that dense graphs were more likely to contain Hamiltonian cycles compared to sparse ones, which often lacked sufficient connectivity. Visualizations created with Matplotlib helped illustrate graph structures and highlight how different graph characteristics influence the presence of Hamiltonian cycles. Overall, the study demonstrates how theoretical concepts in graph theory can be explored through computational methods and showcases Python as an effective tool for modelling and visualizing complex graph problems.

Keywords: Hamiltonian Cycle, Graph Theory, Computational Modelling, NetworkX, Python, Backtracking Algorithm, Graph Visualization, Network Analysis

Adaptive RRI and Priority based scheduling for Mode-2 5G V2X Communication

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The third-generation partnership project (3GPP) published Release 16 in 2020, introducing mode-2 sidelink communication for 5G after successful implementation of LTE V2X mode-4 for communication between vehicles for autonomous Cellular V2X applications in LTE. 5G NR V2X introduced multiple new methods to reduce packet collision, for reliable communication in the absence of a Base Station(BS). Fixed RRIs often fail to adapt to channel congestion, mobility, or hidden terminal scenarios, resulting in excessive collisions, reduced packet reception ratio (PRR), and degraded latency performance under moderate to high vehicular densities. In this study, the authors introduced priority-based Adaptive RRI (ARRI) for resource allocation and better packet reception. In this study, the authors develop a fully decentralized NR-V2X simulation framework that models pathloss-based spatial interference, RSRP-filtered neighbor visibility, distributed sensing windows, periodic multi-slot SCI signaling, and dynamic vehicular priorities. A comparative analysis done using Packet Reception Ratio, Collision Rate, and Latency metrics shows that the new combined method is performing significantly better than the preexisting baseline 5G NR mode-2 and LTE mode-4 V2X. The method is extended with a DRL-based adaptive scheme, which learns a mapping from local observations—including discretized collision buckets, CBR bins, RRI index, priority state, and SCI-derived reservation density—to an optimal RRI decision from a configurable action set. The DRL agent further outperforms the heuristic in several configurations by discovering finer-grained RRI strategies that reduce contention, avoid high-load slots, and select more collision-resilient intervals than any fixed or heuristic policy. Overall, this work presents the first fully parallel, slot-accurate DRL framework for adaptive RRI control in decentralized NR-V2X Mode 2. The results indicate that DRL-based RRI selection is a promising direction for future vehicle-to-vehicle standards and can significantly enhance cooperative awareness in dense and dynamic environments without requiring infrastructure support.

Optimized Audio Embedding in Images using Nature-Inspired Metaheuristics

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In today's digitally connected environment, ensuring the security of sensitive information has become increasingly important. Traditional cryptographic techniques protect content but often reveal the presence of encrypted data, which can attract unwanted attention. Steganography addresses this limitation by concealing the very existence of secret information within a cover medium. Among various steganographic approaches, embedding data within images is particularly effective due to the widespread use and large redundancy of image pixels.

This paper introduces a technique for embedding audio files into images using an enhanced form of the Least Significant Bit (LSB) substitution method. While conventional LSB approaches replace the least significant bits of pixel values directly, they often face challenges related to distortion, limited capacity, and vulnerability to detection. To overcome these issues, the proposed method integrates the Harris Hawks Optimization (HHO) algorithm to intelligently select optimal pixel positions for embedding. HHO is a nature-inspired optimization strategy modeled on the cooperative hunting strategies of Harris hawks. Its exploration–exploitation balance allows it to efficiently navigate the search space and identify embedding positions that minimize visual distortion while maximizing capacity.

The performance of the proposed system is assessed using widely accepted metrics: Peak Signal to-Noise Ratio (PSNR) to measure visual quality, Structural Similarity Index (SSIM) to evaluate perceived similarity between original and stego images, and Mean Square Error (MSE) to quantify pixel-level differences. Experimental results indicate that the integration of HHO significantly improves embedding performance compared to standard LSB and other heuristic-assisted approaches. The images retain high perceptual quality, demonstrating minimal visible alteration even at higher embedding rates. Additionally, the method shows strong robustness against common image processing operations, maintaining the integrity of the hidden audio data. These findings highlight the method's potential for secure multimedia communication where imperceptibility and reliability are essential.

Memristor Crossbar Array – From In-Memory Computing to Neuromorphic Computing

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A memristor is a type of resistance-switching memory device capable of mimicking biological synaptic functions in artificial neural networks. A memristor is an emerging electronic device that provides key features to combine memory and computational elements at the same location to reduce device size and power consumption. Crossbar-based memristors have demonstrated advantages in physically executing vector-matrix multiplication (VMM) and enabling highly power-efficient and area efficient neuromorphic system designs. They have also been extensively explored to replace the traditional von Neumann architecture, allowing processing and storage at the same location, commonly known as in memory computing architecture (IMC). These IMC architecture exploits the inherent switching properties of memristor to realize logic gates.

Due to its high density, low access energy, and nanoscale dimensions, the memristor has been regarded as one of the most promising emerging non-volatile memory technologies. Utilizing memristors to store data is area-efficient due to their ability to store multiple bits per cell. Programming and reading data from memristors are achieved by applying a voltage across the memristor above and below a threshold, respectively. Spiking neuron models for neuromorphic computing or resistive switching models for in memory computing.

Using circuit simulation tools like HSPICE and Cadence Virtuoso to simulate the behaviour of memristor-based systems in both in-memory and neuromorphic computing scenarios. Analysing the effectiveness of memristor-based in-memory systems by comparing them with traditional architectures like von Neumann, in terms of: scalability, energy efficiency, speed.

Memristors can reshape in-memory computing, neuromorphic systems, and bioinspired computing, leading to more efficient, adaptive, and biologically relevant computational systems. Their ability to simultaneously store and process information and their inherent properties that mimic biological processes opens up new frontiers of efficient, adaptive, and scalable computing systems.

Design and Implementation of a Realistic 5G RAN Slicing Simulator for Digital Twin–Driven Control Optimization Using Deep Learning

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The rapid growth of 5G services has created a clear need for networks that can flexibly support very different types of traffic, ranging from high-speed applications to delay-sensitive and large-scale machine communication. However, experimenting on a live network is very costly, restrictive, and highly risky, making it difficult to study how resources should be divided between multiple service types.

To address this challenge, we developed a full 5G network slicing simulation framework using ns 3.46 with the CTTC NR module (4.1.1) that recreates the behaviour of a sliced 5G network. It allows us to recreate radio access setup and control how different services will co-exist. We model resources such as bandwidth allocation, processing capability, and transmission strength as tunable parameters that can be dynamically reassigned to reflect changing user conditions and service loads. The system records key performance indicators including throughput, delay, jitter, loss rate, and slice-level behaviour, enabling a detailed understanding of the impact of each control decision.

Our methodology revolves around repeatedly adjusting slice-level actions and observing how the simulated network responds. This creates a comprehensive dataset reflecting real-world conditions such as user mobility, signal variation, and competition for shared radio resources. We run multiple simulations with varying parameters including the number of UEs per slice, available bandwidth, transmit power, and propagation-loss conditions. These experiments allow us to evaluate how different slicing strategies influence Quality of Service and overall user experience.

Future Work: The data generated by this simulator will serve as the foundation for a 5G Digital Twin. We plan to train an LSTM-based model to predict future network states and a deep neural network to estimate slice-level rewards. Using these two components, we aim to apply deep reinforcement learning to enable autonomous optimisation of control actions, achieving adaptive and self-managed 5G slicing without interacting with a real 5G network or simulation.

Extreme Heavy Rainfall Events in Uttar Pradesh (India) During the 2019 Southwest Monsoon: A case study using Meteorological and Remote Sensing observations

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This paper aims to study the environmental conditions associated with extreme heavy rainfall events over Uttar Pradesh, India, during the southwest monsoon 2019. Observed rainfall data showed that extremely heavy rain fall instances were dominated mainly in the second week of July and the last week of September. Only those categorized as extremely heavy rainfall ($r \geq 204.5$ mm/day) and persisted for two days or more are considered for detailed environmental features. Accordingly, two episodes (10-11 July and 27-30 September) are selected. Analysis showed that the major synoptic systems observed during these episodes were low-pressure areas, cyclonic circulations, and other thermodynamic and dynamic features over the region. This study also highlights rainfall estimates from ERA5 and remote sensing tools like INSAT-3D/3DR satellite data in analyzing EHR events over UP during SWM 2019, showing intense convective activity and high rain rates during these EHR episodes, mainly during 10-11 July. This case study may apply to disaster management, operational forecasts, warning services, water resource management, and other allied disciplines.

Keywords: Heavy rainfall, Southwest Monsoon 2019, Meteorological analysis, Uttar Pradesh

Visual QKD: A Software Platform for Quantum Key Distribution Simulation and Visualization

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Visual QKD is an advanced software-based platform designed to simulate, visualize, and analyse Quantum Key Distribution (QKD) protocols in an intuitive and interactive environment. Developed with the goal of bridging the gap between theoretical quantum communication concepts and real-world implementation, Visual QKD enables researchers, students, and professionals to explore quantum cryptography without requiring complex optical setups or costly hardware. The platform provides a visual interface to model popular QKD protocols such as BB84, E91, and decoy-state variants. Users can design complete QKD systems—from photon sources and polarization states to detectors and error correction modules—through an easy-to-use graphical interface. Real-time visualization of photon transmission, basis selection, and key reconciliation helps users understand the quantum principles underlying secure communication, including superposition, measurement, and quantum randomness. In addition to visualization, Visual QKD offers data generation tools for performance analysis under varying conditions such as noise, loss, and eavesdropping attacks. This feature makes it valuable not only for educational purposes but also for feasibility studies and research proposal preparation. Researchers can export figures, datasets, and simulation results to support scientific publications or project reports. By eliminating the need for physical hardware, Visual QKD democratizes access to quantum communication research and training. It promotes reproducible experimentation, enhances learning outcomes, and supports innovation in the growing field of quantum technologies. Designed and developed at IIT Delhi, Visual QKD represents a step toward making quantum security research more accessible, visual, and collaborative.

Petri Net–ML Based Approach to Vancomycin-Resistant *Enterococcus*

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Antimicrobial resistance (AMR) is a major global health threat, undermining the effectiveness of existing therapies and increasing morbidity and mortality. Addressing this escalating challenge requires innovative and integrative approaches to drug discovery and therapeutic targets. *Enterococcus* species, listed in the Indian Priority Pathogens List (IPPL), are increasingly resistant to vancomycin, a tricyclic glycopeptide antibiotic derived from *Streptococcus orientalis*, is a critical treatment for severe Gram-positive infections caused by methicillin-resistant *Staphylococcus aureus* (MRSA), methicillin-susceptible *S. aureus* (MSSA), and *Streptococcus* and *Enterococcus* species. It acts by inhibiting bacterial cell wall synthesis. Resistance genes, often carried on plasmids and transposons, enable horizontal gene transfer. This study uses Petri net modeling coupled with graph-theoretic analysis to simulate vancomycin resistance pathways. In this framework, places represent biological conditions, transitions represent events, and token distributions depict state changes, which are further explored using reachability analysis. Graph theory concepts, including network centrality, connectivity, and motif detection, are applied to the Petri net-derived networks to identify critical genes, regulatory hubs, and potential points of intervention. Invariant analyses, including t-invariants with MCT sets and t-clusters, reveal conserved functional cycles. Machine learning and data mining further uncover predictive genes and unusual genetic patterns from DBT's IPPL datasets. This integrative framework—combining Petri nets, graph theory, and computational learning—provides a robust approach to understanding, predicting, and potentially mitigating vancomycin resistance in *Enterococcus*.

Keywords: Antimicrobial Resistance, *Enterococcus*, Petri Net Modeling, Graph Theory, Vancomycin Resistance, Machine Learning

Design Study of C-Shaped Antenna With AMC for Wearable IoT Healthcare Devices Application

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This work presents the design and analysis of a compact C-shaped antenna integrated with an Artificial Magnetic Conductor (AMC) structure for wearable IoT healthcare applications. The proposed antenna features an ultra-compact footprint of $30 \times 30 \times 0.7 \text{ mm}^3$, making it suitable for body-worn and space constrained platforms. To enhance its electromagnetic performance in close-to-body conditions, a single-layer AMC surface of $27 \times 27 \times 0.7 \text{ mm}^3$ is employed beneath the radiator. The AMC unit is engineered to provide an in-phase reflection response centered at the 2.2 GHz resonant frequency, ensuring improved impedance matching, reduced surface-wave propagation, and enhanced radiation characteristics when worn on the human body. The miniaturization is primarily accomplished through the implementation of a meandered resonator, which effectively increases the electrical length of the antenna within a compact physical footprint. By folding the resonant path into a meandered structure, the design achieves lower resonant frequencies without expanding the overall size, enabling efficient operation at 2.2 GHz while maintaining the small form factor required for wearable IoT healthcare applications. Performance evaluations indicate that the proposed antenna achieves a realized gain improvement when backed by the AMC surface, demonstrating stable and efficient radiation at the 2.2 GHz operating frequency. The antenna exhibits an isolation level ranging from -18 dB to -25 dB across the operating band, ensuring minimal interference and reliable signal integrity in wearable environments. These results confirm the effectiveness of the AMC structure in enhancing radiation performance, suppressing surface waves, and reducing unwanted coupling, which are critical factors for IoT-based healthcare devices operating in close proximity to the human body. Overall, the proposed antenna offers a promising solution for next-generation implantable medical systems, delivering compact size, enhanced radiation performance, and reliable operation in challenging in-body environments. Its optimized geometry, combined with the AMC backed configuration, ensures efficient communication, reduced tissue absorption, and improved safety-making it well-suited for future biomedical sensing and monitoring applications.

High-Gain Wideband Suspended Equilateral Triangular Patch Antenna with a Capacitive Feed

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In the rapidly evolving domains of wireless communication and radar systems, the demand for antennas capable of delivering both high gain and wideband performance has increased substantially. Conventional microstrip patch antennas are widely recognized for their simplicity, compact size, and ease of integration; however, their ability to simultaneously achieve high gain and wide operational bandwidth remains limited due to inherent design trade-offs. To address these challenges, recent research has focused on innovative antenna structures that enhance electromagnetic performance without compromising compactness. Among these, the suspended equilateral triangular patch antenna with a capacitive feed has emerged as a promising solution.

The equilateral triangular geometry, characterized by its inherent symmetry and compact footprint, supports multiple resonant modes that contribute to broader impedance bandwidth. This geometric advantage, combined with the suspended configuration, effectively modifies the interaction between the patch and the underlying ground plane. As a result, surface wave propagation is reduced, impedance matching is improved, and radiation efficiency is enhanced. The suspended structure also allows for greater control over the effective dielectric constant, enabling improved frequency agility and wider operational bandwidth.

Furthermore, the incorporation of a capacitive feed mechanism provides an additional degree of freedom in impedance tuning. This feeding technique enhances coupling efficiency, facilitates wideband matching, and improves overall radiation characteristics. Together, the suspended patch configuration and capacitive feed synergy enable the antenna to achieve high gain while maintaining compactness and supporting wideband operation.

Overall, the suspended equilateral triangular patch antenna with capacitive feeding represents a robust and innovative design approach capable of meeting the growing performance demands of next generation wireless and radar systems. This work highlights its potential as a competitive candidate for applications requiring high efficiency, wide bandwidth, and compact form factors.

STUDY ON VARIOUS ANTENNA TECHNOLOGIES FOR THE WIRELESS COMMUNICATION

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Antennas play a vital role in modern wireless communication systems, enabling efficient transmission and reception of electromagnetic waves. This project presents a detailed study and comparative analysis of commonly used antennas, focusing on their structural designs, operating mechanisms, S-parameter characteristics, and performance results. The selected antennas include the dipole, monopole, loop, microstrip patch, slot, horn, Yagi–Uda, helical, parabolic reflector, and MIMO antenna. Each antenna is designed using standard electromagnetic simulation tools.

The project evaluates the S11 parameter for each antenna to determine return loss, bandwidth, and impedance-matching efficiency. The S-parameter results show that dipole and monopole antennas provide broad coverage and moderate gain, while microstrip patch antennas offer compact size yet limited bandwidth. Slot and loop antennas exhibit good miniaturization, making them suitable for low profile wireless devices. High-gain antennas such as horn, Yagi–Uda, and parabolic reflectors display strong directional radiation patterns, making them suitable for radar, satellite, and long-range communication systems. Helical antennas provide circular polarization, enhancing communication stability in GPS and satellite links. MIMO antennas show significant improvements in channel capacity, diversity gain, and isolation, supporting advanced 4G and 5G applications.

The study presents a conceptual overview of how different antenna types can be compared based on key parameters such as bandwidth, gain, polarization, and physical size. The discussion, while it provides no specific antenna designs and simulation results, indicates how structural variations typically influence S parameter behaviour and antenna performance. It provides foundational guidelines for selecting appropriate antenna configurations for emerging wireless technologies.

STUDY OF VARIOUS COMPONENTS USED IN MICROWAVE BENCH IN MICROWAVE LABORATORY

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The microwave bench is used to analyze the propagation characteristics, measurement techniques, and performance parameters associated with microwave-frequency electromagnetic waves using a standard laboratory waveguide setup. Microwaves, typically operating in the gigahertz range, show behaviors significantly different from low frequency signals, necessitating specialized components and measurement arrangements. In the microwave bench , a Gunn diode oscillator or a Reflex Klystron tube serves as the primary microwave source, generating continuous-wave signals that propagate through a rectangular waveguide. An isolator is incorporated to prevent reverse power flow and ensure source stability, while a variable attenuator sets the amount of attenuation and enables controlled adjustment of signal amplitude. A precision frequency meter is used to determine the operating frequency of the generated microwaves. Frequency Meter measures the frequency of the signal.

The slotted line section, equipped with a high-sensitivity probe, plays a critical role in detecting the standing wave pattern created by the interaction of forward and reflected waves within the waveguide. By accurately locating the minima of the standing wave, the guided wavelength (λ_g) is experimentally determined. Using this guided wavelength, the operating frequency is calculated through the relation $f = c/\lambda_g$, appropriately corrected for waveguide dispersion. Additionally, measurements of the maximum and minimum detected voltages allow the computation of Voltage Standing Wave Ratio (VSWR), from which the reflection coefficient (Γ) is derived.

Microwave Bench provides a comprehensive understanding of microwave propagation, impedance mismatching effects, attenuation mechanisms, and standing wave phenomena. Microwave Bench study forms a foundation for advanced study in microwave engineering and are vital for applications in radar technology, satellite communication, and high frequency circuit design.

Design Study of Filters in RF Communication

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This work presents a design study and performance evaluation of two essential RF filters, low-pass and band-pass, implemented using microstrip technology for modern wireless communication systems. The band-pass filter (BPF) is realized using a parallel coupled-line configuration with copper microstrip conductors and an Alumina substrate ($\epsilon_r = 9.9$, thickness = 25 mil). The filter is optimized to operate within the 3.3–3.8 GHz band, achieving an insertion loss of approximately 1.053 dB, making it suitable for mid-band 5G applications. The low-pass filter (LPF) is designed using a rectangular patch Compact Microstrip Resonant Cell (CMRC) integrated with a Defected Ground Structure (DGS). Rogers RT5800 substrate with a thickness of 0.381 mm is used, and the total filter size is compact at 45×30 mm. The LPF exhibits a cut-off frequency of 2.4 GHz, offering effective suppression of higher-order harmonics while maintaining a small footprint.

Design of a Multipurpose Vivaldi Antenna for Car Rooftop Applications enabling Vehicle-to-Vehicle Communications

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This study presents an efficient and lowcost rooftop car antenna for the purpose of Car to Car communications and Wi-Fi6 applications. The proposed design is a Vivaldi antenna made by gradually tapering the exponential arms on FR4 substrate of permittivity $\epsilon_r = 4.3$. This antenna is compact sized 90mm * 53mm * 1.57mm which can be mounted on a car roof with the capability of providing modern communication and high speed Wi-Fi services to vehicles. This antenna resonates at 5.8GHz frequency band suitable for V2V communications and Wi-Fi6 applications with a return loss of around -38dB offering an excellent match and a VSWR below 1.5. Additionally, antenna's gain increases with the frequency and has a realized gain of 7.7dB at 5.8 GHz. This antenna has main lobe direction at 91° and 3dB width is 56.7° . This antenna works well for the various applications in 5-6GHz frequency range making it a good choice for rooftop car antennas. The proposed antenna element is designed and simulated in CST studio 2024 version. Furthermore, The 4*1 antenna array, made by arranging these Vivaldi antennas on the mid points of square's edges in cruciform shape provides an omnidirectional radiation pattern. The omnidirectional radiation pattern provides 360° horizontal coverage with a bit downward tilt to maintain reliable V2V communication patterns in all directions. This array antenna can provide far and comprehensive coverage when placed perpendicularly to the vehicle's roof. Beside the V2V application, the operation bands of the antenna are suitable for 5G, Wi-Fi 6, and further communication systems in the future.

Keywords : V2V Communication, WiFi 6, Vivaldi Antenna, Antenna Array, Multipurpose Antenna, Car Roof Top Antennas

Design Study of Miniaturized MIMO Antenna for Implantable Biomedical Applications

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A compact four-port Multiple-Input Multiple-Output (MIMO) antenna tailored for implantable medical devices is presented in this study. The proposed design integrates a meandered resonator patch and a strategically slotted ground structure to achieve significant size reduction without compromising essential electromagnetic performance. With overall dimensions of $6.2 \times 5.35 \times 0.12 \text{ mm}^3$, the antenna is optimized to operate efficiently at the 2.45 GHz ISM band, a frequency widely used in biomedical telemetry applications.

The miniaturization is primarily accomplished through the implementation of a meandered resonator, which effectively lengthens the current path within a limited physical area, while the slotted ground enhances impedance matching and contributes to improved radiation characteristics. To support the high data-rate requirements of contemporary implantable communication systems, a four-port MIMO configuration is incorporated directly on the ground plane. This configuration enhances channel capacity and reliability in lossy biological environments.

Performance evaluations indicate a realized gain of -42.5 dBi , characteristic of deeply implanted antennas operating within tissue-mimicking conditions. The antenna exhibits an isolation level ranging from -16 dB to -20 dB , demonstrating satisfactory inter-port decoupling for biomedical MIMO applications. Additionally, the design achieves a 380 MHz impedance bandwidth, ensuring robust operation across the intended medical communication band.

Overall, the proposed antenna offers a promising solution for next-generation implantable medical systems requiring ultra-compact form factors, reliable multi-channel communication, and efficient operation within the constraints of the human body.

Index Terms— Implantable, meandered resonator, MIMO, ISM (Industrial, Scientific, and Medical), slotted ground

Brain Tumor Image segmentation

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This project addresses the critical task of semantic segmentation of brain tumors from medical images, leveraging deep learning techniques to enhance diagnostic accuracy and treatment planning. We utilized a comprehensive brain tumor image dataset, annotated in COCO format, for training, validation, and testing. The core of our approach involved building a U-Net architecture integrated with transfer learning, employing a ResNet50 backbone pretrained on ImageNet to harness robust feature extraction capabilities. The model was trained using a combined Binary Cross-Entropy and Dice Loss function, optimized with Adam, and monitored for Dice Coefficient, IoU, and accuracy. Data augmentation techniques, including random horizontal/vertical flips and rotations, were applied to improve model generalization. The trained model achieved a Dice Coefficient of 0.8094, an IoU of 0.6821, and an accuracy of 0.9864 on the test set, demonstrating strong performance in accurately delineating tumor regions. These results highlight the potential of transfer learning within U-Net architectures for precise brain tumor segmentation, offering a valuable tool for medical imaging analysis.

Interactive Simulation, AI-Driven Gameplay Systems, and Real Time Procedural Environments

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This project exhibition presents a curated collection of four interactive game systems developed by **Team Ephirium**, the Esports and Game Development Club of the School of Engineering. Designed across different genres and engineering challenges, the portfolio demonstrates practical implementations of artificial intelligence, real-time graphics, procedural generation, and scalable interactive systems using **Unity (C#)** and **Roblox Studio (Lua)**.

The first project, **Zero Horizon**, is a third-person 3D shooter featuring autonomous enemy agents capable of navigation, pursuit, and combat using Unity's NavMesh system. The project highlights behaviour-driven AI, animation control, and integrated combat logic to create responsive and intelligent gameplay dynamics.

The second project, **Young Granny**, is a rapid-development horror game built in under seven hours during India's first AI Game Jam organized by IICS. The project explores accelerated development pipelines using AI-assisted workflows for prototyping, environment creation, and narrative design.

The third project, **Ambient Trip**, is an endless driving experience featuring obstacle-avoidance mechanics and **procedural environment generation**. It demonstrates algorithmic content creation, optimized rendering, and real-time physics for smooth gameplay at extended durations.

The fourth project, **Roblox NPC System**, implements a large-scale interactive NPC network where agents generate **real-time natural-language responses** to player inputs using AI models. This system showcases scalable interaction handling, data routing, and multi-agent simulation within Roblox's Lua based environment.

Together, these projects represent an interdisciplinary approach to modern game engineering— combining AI, simulation, procedural systems, and real-time interaction. The portfolio highlights the potential of interactive digital environments for applications beyond entertainment, including training simulations, rapid prototyping, and experiential learning.

Cross-Modality Brain Image Translation Using Unpaired and Paired Deep Learning Frameworks

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In diagnosing neurological conditions, brain imaging plays a crucial role yet due to practical limitations, clinicians cannot access CT and MRI scan both for every patient. CT imaging is fast and easily available but offers limited insights as it displays soft-tissue contrast, while MRI provides rich anatomical detail but is costlier than CT, consumes more time and is not easily accessible in emergencies or places with low resources. This gap restricts diagnostic precision and motivates the need for intelligent systems capable of synthesizing one modality from another.

In this project, we develop a two-stage deep learning framework to generate MRI-like images from CT scans and vice versa. In stage 1, we employ an unpaired CycleGAN model which enables the system to learn modality translation using large, unaligned datasets. This adversarial and cycle-consistent training allows the model to capture structural patterns and modality-specific characteristics without requiring matched CT-MRI pairs. In stage 2, we refine this translation through paired learning using supervised loss functions. The system enhances structural accuracy, tissue consistency and overall realism of the synthesized images by integrating paired CT-MRI samples.

The system is trained on approximately 10,000 brain CT DICOM images from the RSNA Intracranial Haemorrhage Detection dataset and 10,000 brain MRI DICOMs from the fastMRI dataset, along with a smaller paired dataset for supervised refinement. All images undergo medical-grade preprocessing, including HU windowing for CT, percentile normalization for MRI, skull-preserving cropping and re sizing to 256x256 pixels.

The final model generates anatomically consistent and visually realistic cross-modality translations which demonstrates improved quality after paired fine-tuning. These results demonstrate the potential of AI-based modality synthesis to support clinical decision-making, enhance imaging availability in constrained environments and expand multimodal datasets for medical AI research.

Deep Learning for Facial Scar Restoration Prediction

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This project aims to address a crucial challenge faced by individuals with facial burns or scars who want to visualize post-surgical outcomes before undergoing reconstructive procedures. Usually, there are no preview tools for the patients to have some expectations from the reconstructive surgery, leading to uncertainty regarding surgical results. Providing intuitive visualizations helps patients and clinicians in treatment planning.

We utilize deep learning to develop an automated pipeline capable of generating realistic facial reconstructions from images showing burns or scars. Using state-of-the-art generative models, the system learns to restore facial features while preserving individual identity, providing possible surgical outcome previews that improve communication and decision-making by allowing the patients to have some sort of expectations from the surgery.

The technical methodology consists of several key steps:

- **Synthetic Data Generation:** Starting with a dataset of 7000+ clean facial images, synthetic scars and burns are generated realistically by restricting lesion masks to facial skin regions identified by face-parsing models. This step is necessary as we do not have a publically available dataset of faces with scars and the same faces after restorative procedures (rightfully so as it would be a matter of privacy). We generate scars on the face along with lesion mask, face-skin mask and canny edges from clean images.
- **ControlNet Conditioning:** Facial geometry is preserved by extracting Canny edge maps from clean images, which serve as condition inputs to the generative model; this is required in order to make sure that reconstruction follows the original structure of the face.
- **Model Architecture:** **ControlNet** is a conditional guidance network which is fine-tuned atop a **Stable Diffusion v1.5** inpainting backbone. ControlNet enables precise spatial control of the restoration, maintaining facial contours and preventing undesired global changes.
- **Training Pipeline:** The diffusion model is trained on synthetic data (clean face, face with scar, binary mask of lesion, canny edge map of clean face), learning to inpaint scars based on masks and edge conditions, ensuring identity preservation with realistic texture synthesis.
- **Inference and Realism Controls:** At inference, moderate denoising and guidance scales prevent over-idealized outputs, while masks constrain changes to scarred areas.